

Rishika Patel

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SUMMARY

A motivated Computer Science student specializing in Cybersecurity and Ethical Hacking with hands-on experience in Vulnerability Assessment, Penetration Testing, and Malware Analysis. Proven technical lead skilled in deploying SIEM solutions using the ELK Stack and engineering machine learning models to solve security challenges.

TECHNICAL SKILLS

Languages: C++, Bash Scripting, C, C#, Python, SQL

Security Tools: BurpSuite, Wireshark, Nmap, Metasploit, Nessus, Kali, Hashcat, Splunk, Ghidra

Cloud & Tech: AWS (EC2, Lambda, S3), Blockchain, Linux, Docker, ELK Stack

Specialized: Reverse Engineering, Malware Analysis, Computer Networks, OWASP, OSI Model

PROJECTS

Containerized HIDS & SIEM Pipeline | Python, Docker, ELK stack, Scapy, Nmap | Cybersecurity Project May 2026

- Architected a containerized SIEM using Docker Compose to orchestrate decoupled ELK microservices, optimizing host CPU/RAM overhead by migrating from a heavy Kali VM environment.
- Developed a Python HIDS sensor via Scapy to capture live network traffic, stream JSON alerts over TCP, and validate behavioral rules using simulated Nmap reconnaissance attacks.

Vulnerability Management Platform | Python, SQLite, Network Security, Socket Programming | Cybersecurity Project Jul 2026

- Engineered a modular vulnerability management platform in Python automating network reconnaissance, asynchronous port scanning, CVE signature matching, and HTTP header auditing.
- Architected a relational SQLite database schema to maintain scan state across multi-phase assessments and enforce data deduplication across network targets.
- Developed an automated security reporting engine that aggregates structured database findings into executive-level vulnerability assessment reports.

Carbon Emission Prediction | Python, Scikit-learn | ML Project

Jun-Jul 2025

- Developed a predictive model using Python and Scikit-learn to forecast CO2 emissions based on industrial activity.
- Performed feature engineering and data normalization, achieving high predictive accuracy for sustainability reporting.
- Visualized critical emission drivers to provide actionable insights for environmental impact analysis.

EXPERIENCE

Cyber Security and Ethical Hacking Intern | Tamizhan Skills

Jun – Jul 2025 Remote

- Conducted simulated penetration tests across web applications using Burp Suite and Nmap, identified vulnerabilities and authored remediation reports that reduced potential exploit vectors.

Green Skills with AI Technologies Intern | Edunet Foundation

Jun – Jul 2025 Remote

- Evaluated AI-driven solutions for optimizing environmental data analysis and reporting.

EDUCATION

Vellore Institute of Technology | B.Tech in Computer Science and Engineering (CGPA: 8.84/10)

Bhopal, India 2023 – ongoing

St. Xavier's Convent School | Senior Secondary (XII), ICSE

Lucknow, India 2022

St. Xavier's Senior Secondary School | High School (X), CBSE (91.4%)

Gonda, India 2020

CERTIFICATIONS

- Blockchain and its Applications (NPTEL, IIT Kharagpur)
- The Bits and Bytes of Computer Networking (Coursera)
- Networking Basics & Cyber Threat Management (CISCO)
- AWS Cloud Practitioner & AWS Skill Builder
- Cybersecurity Job Simulation (Mastercard & Deloitte)

ACHIEVEMENTS

- Solved 300+ LeetCode problems and actively competed in Codeforces contests with a 852 contest rating.
- Selected for the DSCI Cyber For Her Hackathon, qualified for Round 2 of Scripted By{Her} 2.0 Meesho Hackathon and the APCSIP-2026 for a Cyber Cell internship.
- Ranked in the top 2% globally on TryHackMe with a 177-day learning streak in penetration testing and digital forensics.
- Placed 355th out of 5,947 global competitors in the TryHackMe Industrial-Intrusion CTF.
- Secured 7th rank nationally in the Shell n'Zen CTF, solving complex web security and reverse engineering challenges.